Linear constraints

Feasible directions

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Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Feasible directions

Most algorithms are iterative. They move from a feasible point in a given direction.
We must make sure that it is possible to generate another feasible point along that direction.

Definition

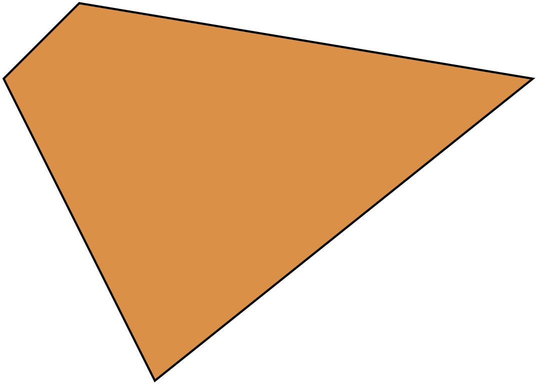
Consider $Y \subseteq \mathbb{R}^n$ be the feasible set and $x \in Y$.
The direction $d \in \mathbb{R}^n$ is feasible in x if $\exists \eta > 0$ such that

$$x + \alpha d \in Y, \forall 0 < \alpha < \eta.$$

Examples

- *Feasible direction at a vertex*
- *Feasible direction from a face*
- *Feasible directions from an interior point*
- *Comment on the fact that there is a maximum step*
- *Examples of infeasible directions*

Examples

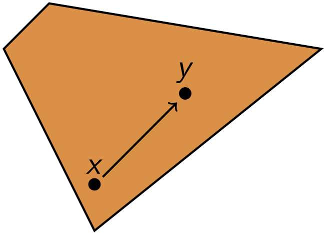


Feasible direction in a convex set

If X is convex, and $x, y \in X$,

$$d = y - x$$

is feasible in x.



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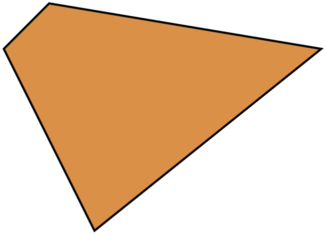

Polyhedron in standard form

$$\mathcal{P} = \{x \in \mathbb{R}^n | Ax = b,\ x \geq 0\}$$

$x^+ \in \mathcal{P}$, $d \in \mathbb{R}^n$, $\alpha > 0$.
Two conditions for d to be feasible

$$b = A(x^+ + \alpha d) = Ax^+ + \alpha Ad = b + \alpha Ad$$

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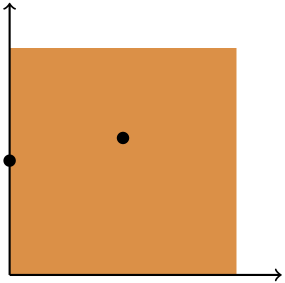
Theorem 3.13: first condition

$$Ad = 0.$$

Polyhedron in standard form

Theorem 3.13: second condition

- If $x_i^+ > 0$, $\forall i$: every direction is feasible.
- If $\exists i$ such that $x_i^+ = 0$, then $d_i \geq 0$.



Summary

A direction is feasible if it is possible to follow it, even a little bit, without leaving the feasible set.
In the case of convex set, finding a feasible direction amounts to identify another feasible point, and select the direction that points to it.
In the linear case, feasible directions should be in the kernel of the equality constraints, and point inward for active inequality constraints.